

Matrix-Structured Parameterization in MCMC for Accelerated Hypertension Prevalence Modeling

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Contents

1	Introduction	3
1.1	Model framework	3
1.2	Parameter estimation	5
1.3	Problems encountered	5
2	Faster MCMC	5
2.1	Induction	5
2.2	Real programming	9
2.3	Performance	9
A	Notation	11
B	Code	12

1 Introduction

Current Hypertension Modeling framework is mainly inspired by the Diabetes Modeling module inputed into old Demographic Epidemiological Model of Singapore (**DEMOS**). And we extend its flexibility for better tuning and integrate our assumption of underlying relationship between systolic and diastolic blood pressure, year effect and Body Mass Index (**BMI**) effect.

1.1 Model framework

Generally speaking, the Hypertension Model is a Bayesian Hierarchical Model. For both log-transformed Systolic value and Diastolic value: the blood pressure for index i at age x with year of birth noted as y_i from g -th gender and e -th stratum under $type \in \{sys, dia\}$ are distributed from a log-normal distribution:

$$bp_i^{g,r,type}(x) \sim LN(BP_i^{g,r,type}(x), \sigma_{type}^2)$$

And the log-normal distribution's mean is generated from a summation of Natural Cubic Spline function (**NC**) of posteriorly-sampled mean blood pressure value at each age knot, time effect and log-scaled BMI trajectory effect:

$$\log(BP_i^{g,r,type}(x)) = \mathbf{NC}(\{\underline{\mu}_i^{g,r,type}\}) + \beta_{type}^{g,r} \cdot (y_i - 1950) + \theta_{type}^{g,r} \cdot \log(BMI_i^{g,r}(x))$$

where $\underline{\mu}_i^{g,r,type}$ are calculated by:

$$\left\{ \begin{array}{l} \mu_{i,18}^{g,r,type} = \mu_{i,30}^{g,r,type} + \varphi_{i,1}^{g,r,type} \cdot (30 - 18) \\ \mu_{i,30}^{g,r,type} = \mu_{i,40}^{g,r,type} + \varphi_{i,2}^{g,r,type} \cdot (40 - 30) \\ \mu_{i,40}^{g,r,type} = \varphi_{i,3}^{g,r,type} \\ \mu_{i,50}^{g,r,type} = \mu_{i,40}^{g,r,type} + \varphi_{i,4}^{g,r,type} \cdot (50 - 40) \\ \mu_{i,60}^{g,r,type} = \mu_{i,50}^{g,r,type} + \varphi_{i,5}^{g,r,type} \cdot (60 - 50) \\ \mu_{i,75}^{g,r,type} = \mu_{i,60}^{g,r,type} + \varphi_{i,6}^{g,r,type} \cdot (75 - 60) \end{array} \right.$$

Individualised blood pressure parameters governing systolic and diastolic blood pressure value at different stages of adulthood ($\varphi_i^{g,r}$) are taken to have multivariate Gaussian distribution over each demographic segment of the population as a whole, with hyper-parameters estimated independently for each race and

gender combination:

$$\underline{\varphi}_i^{g,r} \sim \text{MVN} \left(\begin{pmatrix} M_1^{g,r,sys} \\ \vdots \\ M_6^{g,r,sys} \\ M_1^{g,r,dia} \\ \vdots \\ M_6^{g,r,dia} \end{pmatrix}, \Sigma^{g,r} \right) \triangleq \text{MVN}(\underline{M}^{g,r}, \Sigma^{g,r})$$

To incorporate the stochasticity to our model to handle more complex data, we introduce following hyper prior distributions to our model:

$$\begin{aligned} \underline{M}^{g,r} &\sim \text{MVN}(\underline{\mu}_0^{g,r}, \Sigma_0^{g,r}) \\ \Sigma^{-1} &\sim \text{Wish}(\mathbf{R}_0, 13) \\ \beta_{sys}, \beta_{dia} &\sim U(0.0002, 0.001) \\ \theta_{sys}, \theta_{dia} &\sim U(0.02, 0.05) \\ \sigma_{sys}, \sigma_{dia} &\sim U(0.002, 0.01) \end{aligned}$$

The validity of these hyper prior distribution's parameters has been examined by experts (At least from the BMI paper, they said so). The visualization of the Bayesian Hierarchical Model can refer to the Figure 1 below:

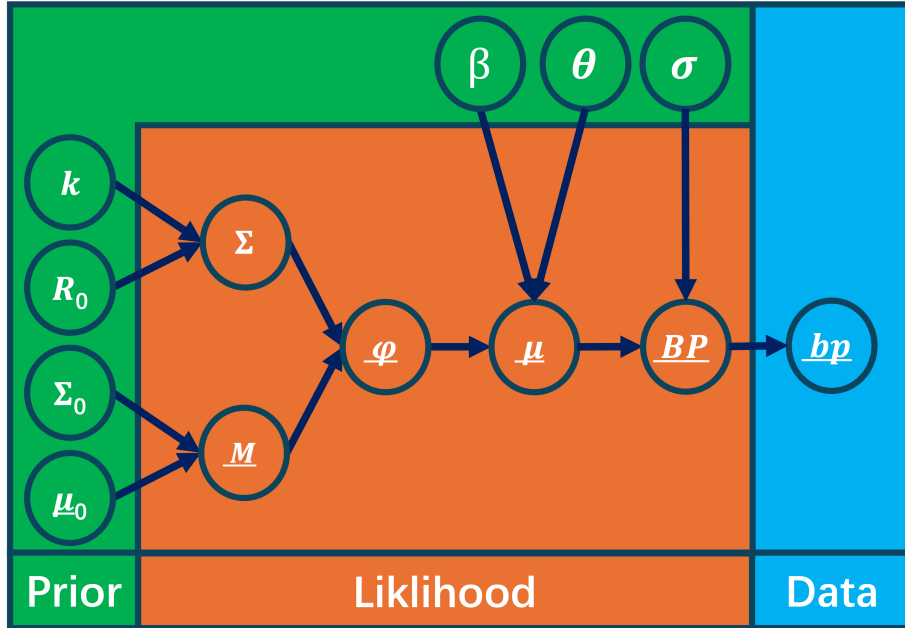


Figure 1: Hypertension Model Illustration

1.2 Parameter estimation

The parameterization of the above hypertension prevalence model is performed by using longitudinal data from Singapore Prospective Study Programme, and after data pre-processing, cohort data used for training can be split into three time points, two time points and one time point case.

Optimal parameter settings are generated from the model based on these data via Monte Carlo Markov Chain (**MCMC**). Then, artificial population is created with similar fraction in gender, race proportion comparing to real Singapore population for simulating blood pressure trajectories to evaluate how hypertension prevalence changes across years, age groups, gender and race.

Data pre-processing and post result analysis are performed in R. The hierarchical model is implemented in JAGS via an interface package named 'R2jags' in R. Generally, for each demographic segment, the program will run 5 parallel chains with 50,000 burn-in steps, 400,000 post burn-in iterations and the number of thinning interval is 350. Gelman-Rubin diagnostic and trace plots are examined for convergence test.

1.3 Problems encountered

During the training of above model, the computing time for each group is quite long. For example, even for minority race group like Malay, the computing is still over 60 hrs, let alone Chinese group. Given that different hyper parameter settings need to be tested in MCMC quite often, shortening computing time of MCMC is necessary.

2 Faster MCMC

After double-checking the model file input into JAGS, we found that in JAGS, it will automatically create deterministic and stochastic nodes for every named variables in the model file, more intermediate variables are created, more time-consuming the model will be to update the parameters. And according the old model file, all parameters in each segment in natural cubic spline function will be created every time in each run of MCMC, which largely slows down the running speed. Considering that natural cubic spline function is all deterministic and linearly transformation, hence, we can write the entire post-stochastic transformation part into a matrix and apply it to those parameters in distributions, i.e., we can express the ultimate $bp_i^{g,r,type}(x)$ in an explicit matrix form without splitting it into several formulas.

2.1 Induction

For simplicity, following inductions will be listed under the context of specific index i , gender g , race r and y_i to be 1950.

Recall from the introduction of model framework above, $\underline{\mu} \in \mathcal{R}^{12}$ can be expressed in a matrix multiplication form:

$$\underline{\mu} = \begin{pmatrix} 12 & 10 & 1 & & & & & & & & & \\ & 10 & 1 & & & & & & & & & \\ & & 1 & & & & & & & & & \\ & & & 1 & 10 & & & & & & & \\ & & & 1 & 10 & 10 & & & & & & \\ & & & 1 & 10 & 10 & 15 & & & & & \\ & & & & & & & 12 & 10 & 1 & & \\ & & & & & & & & 10 & 1 & & \\ & & & & & & & & & 1 & & \\ & & & & & & & & & 1 & 10 & \\ & & & & & & & & & 1 & 10 & 10 \\ & & & & & & & & & 1 & 10 & 10 & 15 \end{pmatrix} \underline{\varphi}$$

Denote the matrix as a big diagonal block matrix, of which diagonal matrix is Q . Therefore, above formula can be re-written as:

$$\underline{\mu} = \begin{pmatrix} Q & \\ & Q \end{pmatrix} \underline{\varphi} \triangleq Q^* \underline{\varphi}$$

Easy to prove that Q has full rank which means it's invertible. Hence, the diagonal block matrix is also invertible. And since $\underline{\varphi}$ is normally distributed, so any linear transformation to normal distribution still lead to normal distribution:

$$\underline{\mu} \sim \text{MVN}(Q^* \underline{M}, Q^* \Sigma Q^{*\text{T}}) \quad (1)$$

After figuring out the expression for μ , we move on to the natural cubic spline part. For example, when $x \in [18, 29]$, according to the spline function, the cubic function to interpolate mean log-scaled systolic blood pressure values in age group $[18, 29]$ can be written as:

$$\log(BP^{sys}(x)) = a_0 + b_0(x - 18) + c_0(x - 18)^2 + d_0(x - 18)^3$$

According to the methodology of **NC**, we know that

$$\begin{cases} a_0 = \mu_1 \\ b_0 = \frac{\mu_2 - \mu_1}{12} - \frac{12}{2} m_0^{sys} - \frac{12}{6} (m_1^{sys} - m_0^{sys}) \\ c_0 = \frac{m_0^{sys}}{2} \\ d_0 = \frac{m_1^{sys} - m_0^{sys}}{6 \times 12} \end{cases}$$

inside the formula, $m_j, j = 1, \dots, 5$ are calculated by

$$A \begin{pmatrix} m_0^{sys} \\ m_1^{sys} \\ m_2^{sys} \\ m_3^{sys} \\ m_4^{sys} \\ m_5^{sys} \end{pmatrix} = 6b$$

where

$$A = \begin{pmatrix} 1 & & & & & \\ 12 & 2(12+10) & & & & \\ & 10 & 10 & & & \\ & & 2(10+10) & 10 & & \\ & & 10 & 2(10+10) & 10 & \\ & & & 10 & 2(10+15) & 15 \\ & & & & & 1 \end{pmatrix}$$

$$b = \begin{pmatrix} 0 \\ \frac{\mu_3 - \mu_2}{10} - \frac{\mu_2 - \mu_1}{12} \\ \frac{\mu_4 - \mu_3}{10} - \frac{\mu_3 - \mu_2}{10} \\ \frac{\mu_5 - \mu_4}{10} - \frac{\mu_4 - \mu_3}{10} \\ \frac{\mu_6 - \mu_5}{15} - \frac{\mu_5 - \mu_4}{10} \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{12} & -\frac{11}{60} & \frac{1}{10} & & & \\ & \frac{1}{10} & -\frac{1}{5} & \frac{1}{10} & & \\ & & \frac{1}{10} & -\frac{1}{5} & \frac{1}{10} & \\ & & & \frac{1}{10} & -\frac{1}{6} & \frac{1}{15} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \underline{\mu}_{1:6} = B \underline{\mu}_{1:6}$$

Obviously, A is strictly diagonally dominated, so A^{-1} exists, i.e., the linear equation has unique solution which is

$$\begin{pmatrix} m_0^{sys} \\ m_1^{sys} \\ m_2^{sys} \\ m_3^{sys} \\ m_4^{sys} \\ m_5^{sys} \end{pmatrix} = 6A^{-1}B\underline{\mu}_{1:6}$$

Therefore, if we consider full vector for both systolic and diastolic version, then

$$\begin{pmatrix} m_0^{sys} \\ \vdots \\ m_5^{sys} \\ m_0^{dia} \\ \vdots \\ m_5^{dia} \end{pmatrix} = 6 \begin{pmatrix} A^{-1} & \\ & A^{-1} \end{pmatrix} \begin{pmatrix} B & \\ & B \end{pmatrix} \underline{\mu} \triangleq 6 C \underline{\mu}$$

Till now, we have proved that each parameter required for spline function can be expressed in a matrix multiplication form of a constant matrix and $\underline{\mu}$. Hence,

back to above condition, when age x is in $[18, 29]$, we can express these parameters into:

$$\begin{cases} a_0 = (1, 0, \dots, 0)\underline{\mu} \\ b_0 = \left[\left(-\frac{1}{12}, \frac{1}{12}, 0, \dots, 0 \right) - (24, 12, 0, \dots, 0)C \right] \underline{\mu} \\ c_0 = (3, 0, \dots, 0)C\underline{\mu} \\ d_0 = \left(-\frac{1}{12}, \frac{1}{12}, 0, \dots, 0 \right) C\underline{\mu} \end{cases} \triangleq \begin{cases} a_0 = \underline{e_{a0}} \underline{\mu} \\ b_0 = \underline{e_{b0}} \underline{\mu} \\ c_0 = \underline{e_{c0}} \underline{\mu} \\ d_0 = \underline{e_{d0}} \underline{\mu} \end{cases}$$

Based on these notations, we can combine it with $\underline{\mu}$'s matrix expression shown previously, the mean log-scaled systolic blood pressure values in age group $[18, 29]$ can be written as:

$$\begin{aligned} \log(BP^{sys}(x)) &= a_0 + b_0(x - 18) + c_0(x - 18)^2 + d_0(x - 18)^3 \\ &= (\underline{e_{a0}} + (x - 18)\underline{e_{b0}} + (x - 18)^2\underline{e_{c0}} + (x - 18)^3\underline{e_{d0}})\underline{\mu} \\ &\triangleq R_{18-29}^{sys}(x)\underline{\mu} \end{aligned} \quad (2)$$

According to (1) and (2), $\forall x \in [18, 29]$ and $x \in \mathbb{Z}$:

$$\begin{aligned} \log(BP^{sys}(x)) &\sim \text{MVN} \left(R_{18-29}^{sys}(x)Q^*\underline{M}, R_{18-29}^{sys}(x)Q^*\Sigma(R_{18-29}^{sys}(x)Q^*)^T \right) \\ &\triangleq \text{MVN} \left(W_{18-29}^{sys}(x)\underline{M}, W_{18-29}^{sys}(x)\Sigma(W_{18-29}^{sys}(x))^T \right) \end{aligned} \quad (3)$$

Similarly, we can derive the expressions for the other age ranges. For the k -th ($k \in [0, 4]$) age range, the systolic coefficients are given by

$$\begin{cases} a_k = \mu_{k+1} \\ b_k = \frac{\mu_{k+2} - \mu_{k+1}}{h_k} - \frac{h_k}{2} m_k^{sys} - \frac{h_k}{6} (m_{k+1}^{sys} - m_k^{sys}) \\ c_k = \frac{m_k^{sys}}{2} \\ d_k = \frac{m_{k+1}^{sys} - m_k^{sys}}{6 \times h_k} \end{cases}$$

$$\begin{cases} a_k = (0, \dots, k, \dots, 0)\underline{\mu} \\ b_k = \left[\left(0, \dots, -\frac{1}{h_k}, \frac{1}{h_k}, 0, \dots, 0 \right) - \left(0, \dots, \frac{h_k}{3}, \frac{h_k}{6}, \dots, 0 \right) 6C \right] \underline{\mu} \\ c_k = (0, \dots, 3, 0, \dots, 0)C\underline{\mu} \\ d_k = \left(0, \dots, -\frac{1}{h_k}, \frac{1}{h_k}, 0, \dots, 0 \right) C\underline{\mu} \end{cases} \triangleq \begin{cases} a_k = \underline{e_{ak}} \underline{\mu} \\ b_k = \underline{e_{bk}} \underline{\mu} \\ c_k = \underline{e_{ck}} \underline{\mu} \\ d_k = \underline{e_{dk}} \underline{\mu} \end{cases}$$

and similarly for diatolic coefficients.

Equations (2) and (3) can be generalized to the other age ranges. Eventually, express both log-scaled systolic and diastolic blood pressure values from [18, 75] and conform them in a 116×1 vector. Then the blood pressure vector is able to be written as a multivariate normally distributed random vector:

$$\log \begin{pmatrix} BP^{sys}(18) \\ \vdots \\ BP^{sys}(75) \\ BP^{dia}(18) \\ \vdots \\ BP^{dia}(75) \end{pmatrix}_{116 \times 1} \sim \text{MVN}(\underline{W}\underline{M}, \underline{W}\Sigma\underline{W}^T)$$

where,

$$\underline{W} = \begin{pmatrix} W_{18-29}^{sys}(18) \\ \dots \\ W_{60-75}^{sys}(75) \\ W_{18-29}^{dia}(18) \\ \dots \\ W_{60-75}^{dia}(75) \end{pmatrix}_{116 \times 12}$$

2.2 Real programming

According to the induction 2.1, we can now express the blood pressure trajectory in a matrix multiplication form and get rid of all redundant intermediate variables and computations. And in the updated JAGS model file, there are three steps:

- Sample $\underline{M}$ and Σ^{-1} from pre-specified prior distributions
- Sample $\underline{\varphi}$ from the multivariate normal distribution whose mean and covariance matrix are $\underline{M}$ and Σ^{-1}
- Calculate the log-scaled blood pressure value with constant matrix $\underline{W}$ and sampled $\underline{\varphi}$:

$$\begin{aligned} \log(BP^{sys}(x)) &= W[x - 17,]\underline{\varphi} \\ \log(BP^{dia}(x)) &= W[x + 41,]\underline{\varphi} \end{aligned}$$

2.3 Performance

With number of parallel chains to be 5, number of burn-ins to be 50,000, number of post burn-in iterations to be 400,000, number of thinning intervals to be 350 and same prior parameters input for Male Malay, we run both old and Faster MCMC model under same desktop Intel Core 4 CPU 2.83 GHz 7.6GB RAM, it takes only 5 hrs for Faster MCMC while old version takes over 60 hrs. And then by examining the convergence of MCMC via Gelman-Rubin diagnostic and

trace plots, Faster MCMC also outperforms older one. Hence, it justifies the validity of the improvement.

A Notation

Notation	Meaning
$i = 1, \dots, N$	individual index in the cohort, N is the total population size in cohort data
$y_i \in \mathcal{R}$	individual's unique year of birth in the cohort data
$x \in [18, 75]$	individual's recorded age in the cohort data
$g = 1, 2$	gender of an individual, 1 for Male and 2 for Female
$r = 1, 2, 3, 4$	race of an individual, 1 for Chinese, 2 for Malay, 3 for Indian and 4 for Other
$type \in \{sys, dia\}$	type of two common different blood pressure value measures: systolic and diastolic
$\beta_{sys}, \beta_{dia} \in \mathcal{R}$	time effects for measuring how individual's year of birth influence their blood pressure trajectories across life time for both types of blood pressure measures respectively
$\theta_{sys}, \theta_{dia} \in \mathcal{R}$	measuring how individual's BMI trajectory affect their blood pressure trajectories throughout their lifetime for both types of blood pressure measures respectively
$\sigma_{sys}, \sigma_{dia} \in \mathcal{R}$	measuring the dispersion of cohort blood pressure value's variance around the mean trend for specific stratum's systolic and diastolic case respectively
$\underline{\varphi}_i^{g,r} \in \mathcal{R}^{12}$	the vector is combined by $\underline{\varphi}_{1:6}^{g,r,sys}$ and $\underline{\varphi}_{7:12}^{g,r,dia}$. They represent the slope of a specific gender + race group's baseline for systolic and diastolic blood pressure trajectory within age group 18-29, 30-39, 40-49, 50-59 and 60-75 respectively. Except for φ_3, φ_9 which stand for the blood pressure value at age 40
$\underline{\mu}^{g,r} \in \mathcal{R}^{12}$	$\underline{\mu}_{1:6}^{g,r} = (\mu_{18}^{g,r,sys}, \dots, \mu_{i,75}^{g,r,sys})^T$ represent the unique systolic blood pressure value at given age for a specific individual in the cohort data and so do $\underline{\mu}_{7:12}^{g,r} = (\mu_{18}^{g,r,dia}, \dots, \mu_{75}^{g,r,dia})^T$ for diastolic case
$bp_i^{g,r,type}(x)$	an individual's real systolic and diastolic blood pressure value with its unique index i , gender g , race r at a specific age x from cohort data

B Code

```
# Model systolic and diastolic BP together
for (i in 1:N.3) {
  # Define prior distributions for systolic and diastolic BP
  for (j in 1:3) {
    SYS.3[i,j] ~ dnorm(mu.sys.3[i,j], tau.sys)
    DIA.3[i,j] ~ dnorm(mu.dia.3[i,j], tau.dia)
    mu.sys.3[i,j] <- inprod(W[AGE.3[i,j]-17,], phi.3[i,])
    + beta.sys*(YOB.3[i]-1950)
    + theta.sys*log(BMI.3[i,j])
    mu.dia.3[i,j] <- inprod(W[AGE.3[i,j]+41,], phi.3[i,])
    + beta.dia*(YOB.3[i]-1950)
    + theta.dia*log(BMI.3[i,j])
  }
  phi.3[i,1:12] ~ dmnorm(M[], Omega[,])
}

# Likelihood for the second set of data with two time points
for (i in 1:N.2) {
  for (j in 1:2) {
    SYS.2[i,j] ~ dnorm(mu.sys.2[i,j], tau.sys)
    DIA.2[i,j] ~ dnorm(mu.dia.2[i,j], tau.dia)
    mu.sys.2[i,j] <- inprod(W[AGE.2[i,j]-17,], phi.2[i,])
    + beta.sys*(YOB.2[i]-1950)
    + theta.sys*log(BMI.2[i,j])
    mu.dia.2[i,j] <- inprod(W[AGE.2[i,j]+41,], phi.2[i,])
    + beta.dia*(YOB.2[i]-1950)
    + theta.dia*log(BMI.2[i,j])
  }
  phi.2[i,1:12] ~ dmnorm(M[], Omega[,])
}

# Likelihood for the second set of data with one time point
for (i in 1:N.1) {
  SYS.1[i] ~ dnorm(mu.sys.1[i], tau.sys)
  DIA.1[i] ~ dnorm(mu.dia.1[i], tau.dia)
  mu.sys.1[i] <- inprod(W[AGE.1[i]-17,], phi.1[i,])
  + beta.sys*(YOB.1[i]-1950)
  + theta.sys*log(BMI.1[i])
  mu.dia.1[i] <- inprod(W[AGE.1[i]+41,], phi.1[i,])
  + beta.dia*(YOB.1[i]-1950)
  + theta.dia*log(BMI.1[i])
  phi.1[i,1:12] ~ dmnorm(M[], Omega[,])
}

# Precision (inverse of variance) for systolic and diastolic BP
tau.sys <- 1/sigma.sys^2
tau.dia <- 1/sigma.dia^2

# Standard deviations for the residuals
sigma.sys ~ dunif(0.002, 0.01)
```

```

sigma.dia ~ dunif(0.002, 0.01)

M[1:12] ~ dmnorm(mu0[],prec0[,])

Omega[1:12,1:12] ~ dwish(R0,13)

S.phi[1:12,1:12] <- inverse(Omega[,])

beta.sys ~ dunif(0.0002, 0.001)
beta.dia ~ dunif(0.0002, 0.001)
theta.sys ~ dunif(0.02,0.05)
theta.dia ~ dunif(0.02,0.05)

```